Rayan Kotowaroo

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Summary of Qualifications

- Fourth-year Mechanical Engineering student at Toronto Metropolitan University with experience in process optimization, data analysis, and structured problem solving within multidisciplinary engineering environments
- Proficient in Microsoft Excel and Power BI for data analysis, KPI tracking, and development of structured reporting tools to support engineering decision-making
- Experience in process documentation, workflow analysis, and project coordination, including task planning, progress tracking, and cross-functional collaboration
- Applied analytical methods to evaluate system performance, identify inefficiencies, and support process improvement initiatives in engineering projects
- Familiar with process and quality engineering concepts, including Lean Six Sigma methodologies (DMAIC, PFMEA, 8D), supporting structured problem-solving and continuous improvement
- Strong communication and organizational skills with bilingual proficiency (English/French), supporting clear documentation, reporting, and collaboration in fast-paced environments

Engineering Experience

Project Manager — ***Metropolitan Aerospace Rocket Society (MARS)*** Sep 2025 – Dec 2025

- Led a 6-member multidisciplinary team by planning timelines, assigning tasks, and tracking progress to deliver a high-power rocket under strict technical and competition constraints
- Coordinated subsystem integration (propulsion, avionics, recovery) by managing cross-functional workflows and ensuring alignment across design, manufacturing, and testing phases
- Monitored project milestones and performance metrics, maintaining accountability for deliverables and supporting structured reporting of project progress

Industrial Fluid System Optimization & Pump Selection Mar 2026 – Apr 2026

- Performed hydraulic analysis (Darcy–Weisbach, TDH, NPSH) using Excel and Power BI to evaluate system performance (flow rate, head loss, efficiency) and support data-driven design decisions
- Compared multiple system configurations by analyzing trade-offs between efficiency and cost, reducing total dynamic head from 60.9 m → 31.4 m and identifying the optimal design
- Applied Lean Six Sigma methodologies (DMAIC, PFMEA, 8D) to identify inefficiencies, standardize evaluation methods, and enhance system reliability and reduce operational risk through structured problem-solving
- Documented system configurations, analysis methods, and performance results in a structured format to support process standardization and engineering reporting

Sustainable Mobile Shelter Sep 2024 – Dec 2024

- Developed and evaluated multiple design concepts under real-world constraints (mobility, cost, sustainability), using Excel to compare performance trade-offs and support engineering decisions
- Created structured documentation and tracked design iterations to support project planning, workflow consistency, and reporting across development stages
- Applied Lean and 5S principles to improve workflow efficiency and support process standardization during design and evaluation

Bar Linkage Robotic Gripper Sep 2025 – Dec 2025

- Evaluated system performance using precision measurement tools (CMM), identifying deviations and supporting quality control and validation processes
- Performed iterative design improvements by analyzing performance issues and implementing adjustments to enhance system reliability and consistency

Work Experience

Swim Instructor & Front Desk Coordinator — ***Olympian Swimming School*** Jan 2021 – Present

- Instructed children in swimming techniques and water safety, adapting instruction methods to different skill levels to improve performance and learning outcomes
- Managed front-desk operations including scheduling, customer coordination, and record-keeping, maintaining accurate tracking of attendance, payments, and bookings using structured data organization methods
- Supported daily operations by ensuring consistent documentation, efficient workflow coordination, and clear communication in a fast-paced customer-facing environment

Education

B.Eng. – Mechanical Engineering — ***Toronto Metropolitan University*** Apr 2028 | Toronto, ON

Relevant Courses: Fluid Mechanics I & II, Thermodynamics I & II, Heat Transfer, Manufacturing Fundamentals, Statistics, Engineering Economics, Mechanics of Machines